

UNIT 2

MATERIALS CHANGING SHAPE

Learning objectives:

- Understand how heating and cooling change the shape or state of materials.
- Explore how stretching and bending affect materials.
- Learn real world examples of materials changing shape (e.g., melting, freezing, stretching).
- Recognize the importance of shape changing materials in daily life.
- Apply knowledge through hands on activities like melting and freezing experiments.

Materials all around us can change their shape in different ways. We see examples of this in our everyday life, like when ice melts into water or when we stretch a rubber band. This is called **changing shape**. Let's learn more about how materials change their shape when they are heated, cooled, stretched, or bent.

2.1 Heating and Cooling

Heating:

Heating a material means making it warmer. When materials are heated, they can change their shape or even become a different kind of material.



For example:

Metal Rods:

When a metal rod is heated, it can bend or become softer, which is why blacksmiths heat metal to shape it into tools or other objects.

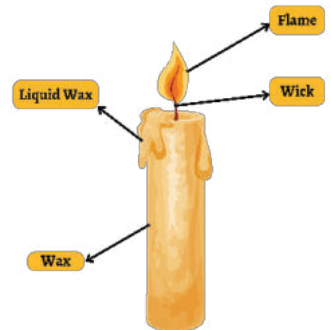


Wax: Wax is soft when it's warm. When you light a candle, the heat makes the wax melt into a liquid.

Cooling: Cooling a material means making it colder. Cooling can make some materials become hard or change their state.

Wax: After a candle is blown out, the melted wax cools and hardens again into a solid.

Water: When water is cooled, like in a freezer, it turns into ice.



2.2 Stretching and Bending

Stretching:

Some materials can stretch without breaking. Stretching means pulling on a material to make it longer or wider.

For example:

Rubber Bands:

Rubber bands can stretch and then go back to their original shape when you let it go. This is because rubber is a flexible material.

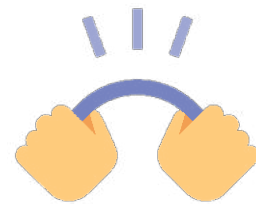
Elastic:

The elastic in clothes stretches so it can fit us comfortably.



Bending:

Bending means changing the shape of a material by pressing or pushing it. Some materials can bend easily, while others cannot.



For example:

Clay:

Clay can be bent and shaped into different forms, like a pot, cup, or sculpture. It's easy to mold because it's soft and flexible.



Plastic:

Some types of plastic can bend a little but return to their original shape. Other types of plastic will stay in the shape you bend them into.



2.3 Examples of Changing Shape

1.Melting:

Melting happens when something solid turns into a liquid because it's heated.

Example:

Ice melts into water when it's left in a warm place. We see this in summer when ice cubes melt in a drink.



2.Freezing:

Freezing is when a liquid becomes solid because it's cooled down.



Example:

Water turns into ice when placed in a freezer. Freezing keeps water in a solid shape, which is why ice cubes stay in one shape until they melt.

3.Boiling:

Boiling happens when a liquid is heated to a very high temperature and turns into a gas (steam).



Example:

When we boil water in a kettle, it gets so hot that it bubbles and releases steam. This steam is water in a gas form.

Why is Changing Shape Important?

Understanding how materials change shape helps us in our daily lives. It shows us:

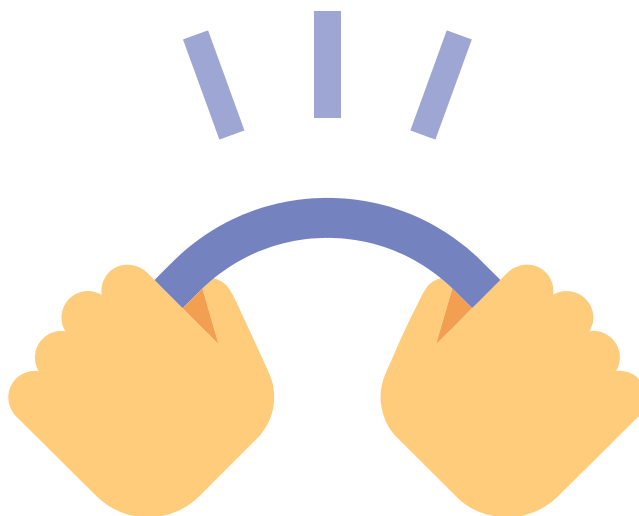
- How to use different materials in different ways.
- Why some materials are better for certain jobs, like using rubber for stretchy things and metal for strong objects.
- How things around us react to temperature changes.

This knowledge helps scientists and engineers design tools, buildings, and items we use every day, like cooking utensils, toys, and clothes.



Fill in the blanks.

1. When a material is heated, it can change its _____.
2. When water is placed in a freezer, it changes into _____.
3. Stretching means pulling on a material to make it _____ or wider.
4. Boiling happens when a liquid is heated to a very high temperature and turns into _____.
5. Rubber is a _____ material, which is why it can stretch and return to its original shape.



Tick (✓) for True and cross (x) for False statements.

1. Wax becomes soft and melts when it is heated. ()

2. Metal rods cannot be bent even when they are heated. ()

3. Freezing turns a liquid into a gas. ()

4. Rubber bands can stretch without breaking. ()

5. Boiling water produces steam, which is water in gas form. ()



Multiple Choice Questions.

1. What happens when wax is cooled after being melted?

- a) It turns into gas
- b) It becomes hard again
- c) It stays liquid
- d) It becomes a powder

2. Which material can easily be stretched?

- a) Metal rod
- b) Rubber band
- c) Glass
- d) Plastic bottle

3. When water is heated to a very high temperature, it:

- a) Freezes
- b) Becomes ice
- c) Boils and turns into steam
- d) Melts

4. What is an example of a material that can be bent and molded into shapes ?

- a) Hard
- b) Soft
- c) Rigid
- d) Transparent



5. Freezing happens when:

- a) A solid turns into a liquid
- b) A liquid turns into a solid
- c) A liquid turns into a gas
- d) A gas turns into a liquid

6. Why do blacksmiths heat metal rods?

- a) To make them melt
- b) To bend and shape them into tools
- c) To break them easily
- d) To turn them into ice



Test Your Understanding

Q1: What happens to a rubber band when you stretch it and then let it go?

Ans: _____

Q2: Give one example of a material that changes when it is heated and one that changes when it is cooled?

Ans: _____

Q3: Describe what boiling does to water and what form it changes into?

Ans: _____

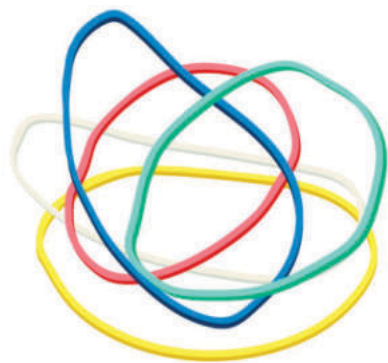
Q4: Why is it important to understand how materials change shape?

Ans: _____



Q5: What happens to water when it is placed in a freezer?

Ans: _____



Critical Thinking

When you press on a clay ball, it changes shape. Why do some materials like clay stay in the new shape, but others, like a rubber band, go back to their old shape?





Exploring Melting and Freezing



ACTIVITY

Understand how materials change state when heated and cooled.

Materials Needed

Ice cubes
Spoon
A bowl of water
Freezer

Instructions:

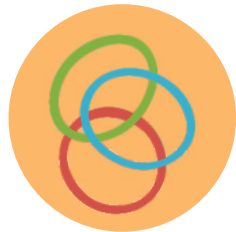
1. Place an ice cube in the warm water and observe what happens. Record how long it takes to melt.
2. Pour water into a cup and place it in the freezer for a few hours. Check to see how the water has changed after freezing.



ACTIVITY

Material That Change Shape

Look at the objects below. Tick the ones that can change shape easily by twisting, bending, squashing, or stretching.

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